VIT | Vidyalankar Institute of Technology

CarWise

Used Car Price Prediction & Market Analysis

Smarter Insights. Fairer Used-Car Pricing.

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PROJECT GUIDE
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ACADEMIC YEAR
2026–2027

PROBLEM STATEMENT

Used-car prices depend on vehicle age, mileage, brand, fuel type, transmission, seller type and ownership history.

Buyers and sellers struggle to find a fair price by intuition alone.

CarWise uses data mining and machine learning to estimate selling prices and reveal market patterns.

OBJECTIVES

— Clean and preprocess used-car listing data.
— Analyse pricing trends and attribute relationships.
— Segment vehicles using K-Means clustering.
— Compare regression models for price prediction.
— Deploy an interactive Streamlit application.

METHODOLOGY

01 **CarDekho Dataset**
02 **Data Cleaning & Feature Engineering**
03 **Exploratory Data Analysis**
04 **K-Means Clustering**
05 **Regression Training & Evaluation**
06 **Streamlit Application**

4,340	763	3,577	7	80/20
original records	duplicates removed	cleaned records	input features	train / test split

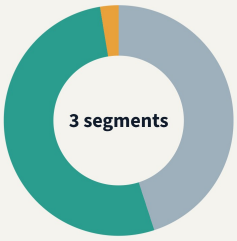
MODEL COMPARISON — TEST SET

Model	Test R^2	Test MAE
Linear Regression	0.541	₹1,79,643
Decision Tree	0.310	₹1,95,085
Random Forest ★	0.589	**₹1,57,686**

RF Test MAE	RF Test RMSE	RF Test R^2
₹1,57,686	**₹3,63,871**	**58.9%**

Random Forest performed best of the three models on held-out test data. R^2 is the coefficient of determination, not a guaranteed prediction accuracy.

MARKET SEGMENTS (K-MEANS)



Budget Daily Drivers
45.0% of records · avg ₹2,50,929

Optimal Commuters
52.4% of records · avg ₹5,53,302

Premium Low-Mileage
2.6% of records · avg ₹27,34,397

KEY INSIGHTS

Vehicle age is associated with a decline in selling price.

Top Random Forest features by importance:

Transmission	24.8%
Brand	24.0%
Car age	22.2%

Random Forest had the lowest test MAE of the three models evaluated.

APPLICATION & STACK

Streamlit app with four modules: **Price Predictor**, **Market Analysis**, **Car Segmentation** and **Model Insights**.

Python · Pandas · NumPy · Scikit-learn · K-Means · Random Forest · Streamlit

CONCLUSION

CarWise combines data preprocessing, exploratory analysis, K-Means clustering and regression modelling in an interactive application to support data-driven used-car price estimation. Random Forest performed best on the test set among the evaluated models, although further tuning and richer vehicle data could improve reliability.